

ALPACA AI TRADING AGENTS HACKATHON — OPTIONS ALPHA

AllPath

trading agent

One kickoff conversation. Then it researches, trades stocks & options through Alpaca's MCP server — and *rewrites its own strategies every night.*

github.com/dukesky/allpath-trading-agent · trading.all-path.com · self-hosted · open source · MIT · 2,368 tests

THE PROBLEM

Analysis bots can't trade. Trading bots can't *think*.

LLM TRADING PROJECTS

Stop at "here's my analysis"

Great research, zero execution. No broker, no risk model, no accountability for whether the idea actually made money.

ALGORITHMIC FRAMEWORKS

Execute but can't reason

Fast and deterministic — but they can't read the news, question a thesis, remember what went wrong last month, or rewrite themselves.

ALLPATH

Closes the loop

An agent that researches like an analyst, executes like a machine, journals every decision, and evolves its own strategies nightly — inside guards no model can bypass.

Three loops, one process

ON DEMAND

Conversation

Web, Telegram, or CLI. The agent interviews you, researches live data, and drafts strategies as readable YAML — thesis in prose, rules as code. Anything touching money becomes an approval card.

EVERY 30 MIN · ZERO LLM COST

Sentinel

Plain code evaluates every rule against live quotes. Hard rules execute with no model in the path; soft rules wake a cheap review agent to research before deciding.

NIGHTLY · AFTER CLOSE

Reflection

A bounded agent session reviews the day against every thesis, writes lessons to curated memory, re-arms spent triggers, and revises its own strategies — as guarded diffs.

rule fires → contract selected via Alpaca MCP → risk gate (deterministic) → order → journal → push receipt → nightly reflection

The LLM writes *intent*. Code picks the contract.

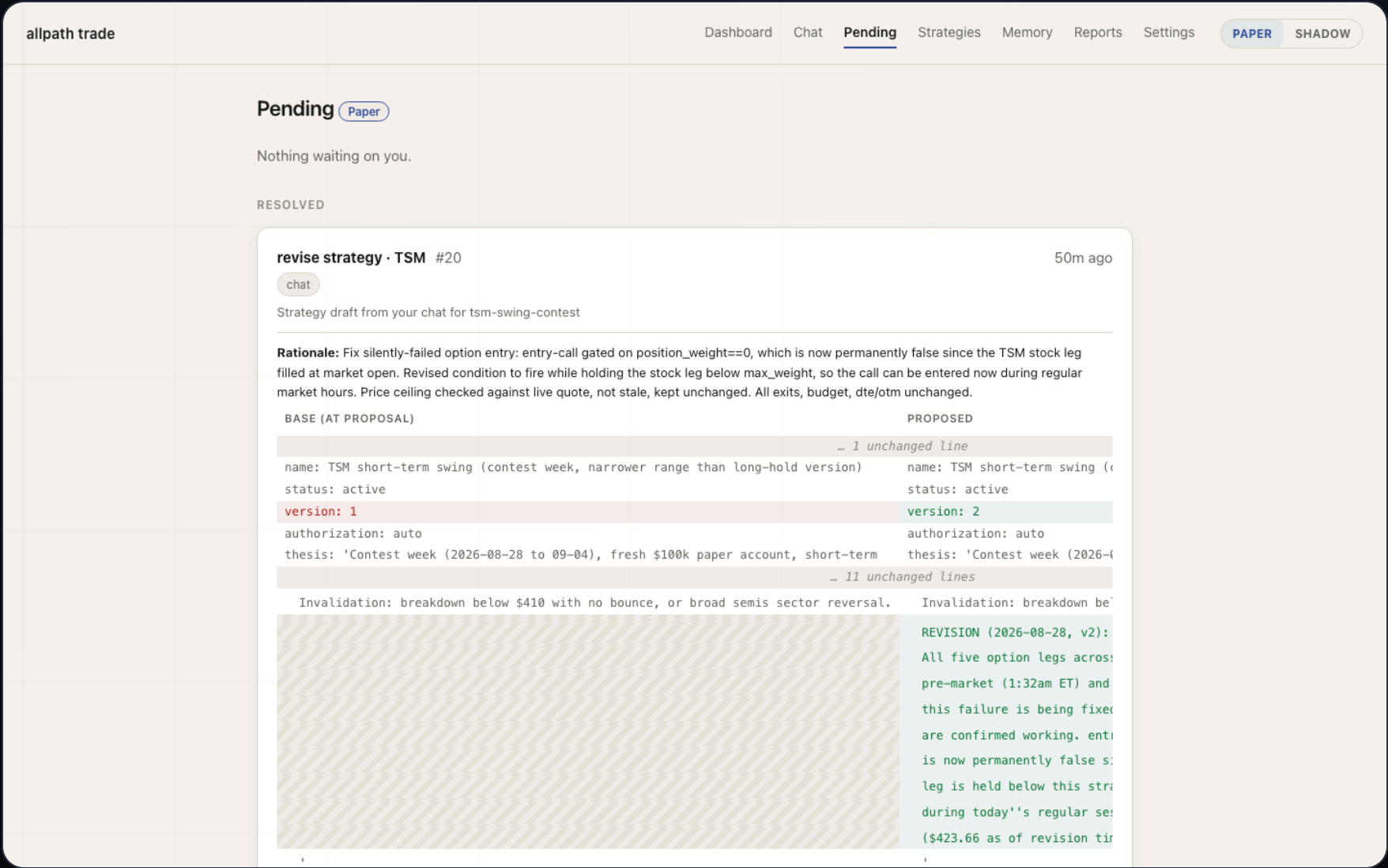
01	Agent writes a rule in its strategy	<code>buy_call \$1500 dte>=5 otm=3%</code>
02	Sentinel trigger → option chain via MCP	<code>get_option_contracts · expirations ≥ today+5, calls only</code>
03	Deterministic contract selection	<code>nearest expiry · strike closest to spot × 1.03 · qty = [\$1500 / (ask × 100)]</code>
04	Risk gate — plain code, non-bypassable	<code>premium ≤ per-order cap · options exposure ≤ 10% equity · cash reserve · daily cap</code>
05	Order placed through MCP, journaled, notified	<code>place_option_order · buy_to_open · market · day</code>

- `managed MCP subprocess · respawn + timeouts`
- `DTE ≤ 1 expiry sweep – never rides into assignment`
- `no order id in response ⇒ journaled as error, never “submitted”`

It found its own bug — *and fixed it.*

Every night, reflection reviews the day against each strategy’s thesis, writes lessons into **curated markdown memory** (profile · strategies · stock dossiers · lessons), and proposes revisions as **diffs** — version-checked, byte-exact against the file they were written for, and frozen out of touching authorization.

Day one: option entries gated on “no position yet” were permanently disabled once the stock legs filled. Told what happened, the agent rewrote its own entry conditions across four strategies, kept the price ceilings after re-verifying them against live quotes, and re-armed the rules. The calls filled the same session.



The LLM can *never* bypass your limits

- ✓ **One path to the broker** — the executor is the only submitting code; the model can only produce intents
- ✓ **Deterministic risk gate** — order value, position weight, options exposure, cash reserve, daily cap
- ✓ **15% drawdown circuit breaker** — halts new risk on every channel; exits still execute
- ✓ **Risk reduction is never blocked** — stop-losses and option closes are exempt from value caps
- ✓ **Expiry sweep** — options with ≤ 1 day to expiry are closed automatically
- ✓ **Market-hours discipline** — closed-market option triggers stay armed instead of burning
- ✓ **No self-promotion** — reflection can never flip a strategy to auto; only a human can
- ✓ **Stale writes refused** — byte-exact base checks and strict version increase on every revision
- ✓ **Honest failure** — a response without an order id journals an error and alerts, never a phantom fill
- ✓ **Hard rules survive outages** — stop-losses run with no LLM in the path

Zero-intervention, fully journaled

\$100k

FRESH PAPER ACCOUNT · ONE
KICKOFF CHAT, THEN HANDS-OFF

5

STRATEGIES THE AGENT DRAFTED
ITSELF — STOCK + OPTION LEGS,
PAIRED EXITS

9

FILLS ON DAY ONE · 4 STOCK
LEGS + 5 OPTION LEGS VIA
ALPACA MCP

1/night

REFLECTION REPORT WITH FULL
TOOL-CALL TRANSCRIPT · SELF-
REVISIONS APPLIED

09:32 ET	BUY 13.24 NVDA @ 226.45	sentinel · entry-momentum · risk gate passed	filled
11:46 ET	BUY 13× NVDA 09/02 230C @ 1.08	contract selected via Alpaca MCP	filled
11:46 ET	BUY 5× CRM 09/04 252.5P @ 2.79	pair hedge — now offsetting the tech pullback	filled · +\$405
11:45 ET	BUY 1× MU 09/02 965C	risk gate: daily trade limit (10/10)	rejected — re-armed by the agent

Every trade, every skip, every LLM decision, every nightly report — journaled. *The audit trail is the demo.*